# What Drives Total Medical Expenses?

## An OLS Analysis of IPUMS Health Survey Data

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**Abstract**

This paper asks which observable individual characteristics predict total annual medical expenses, using a cross-section of 2,016 adults from the IPUMS Health Surveys. Contrary to the initial hypothesis that higher income is associated with higher spending, income alone has a small negative association with medical expenses, about $34 less per $1,000 of income, and explains less than 0.5 percent of their variation. In multiple regressions with demographic and utilization controls, the number of doctor visits and age are the dominant predictors (about $939 per visit and $95 per year of age in the baseline specification), while an interaction between income and doctor visits is not significant. Marital status, less-than-high-school education, and insurance status are significant only at the 10 percent level, and gender and race are not significant. The model explains 17 percent of the variation in spending. A RESET test rejects the linear functional form, and a Breusch-Pagan test does not reject homoskedasticity at the 5 percent level, although residual plots suggest that the variance rises with doctor visits.

*JEL classification:* C21, I10, I11
*Keywords:* medical expenditures; health care utilization; income; ordinary least squares; specification tests

## 1 Introduction

Medical expenses vary enormously across individuals. Some people spend nothing in a given year, while others spend tens or hundreds of thousands of dollars. This paper asks which observable individual characteristics predict total annual medical expenses, and how much of the variation in spending those characteristics can explain.

The starting hypothesis was that higher annual income would be associated with higher medical expenses, on the reasoning that higher-income individuals can afford more doctor visits, more specialists, and more out-of-pocket care. The data do not support it. In a sample of 2,016 adults from the IPUMS Health Surveys, the correlation between income and medical expenses is weak and negative ($r = -0.07$), and income alone explains less than half of one percent of the variation in spending.

Adding demographic and utilization controls changes the picture. The number of doctor visits and age are the dominant statistically significant predictors, while income keeps a small negative coefficient. An interaction between income and doctor visits, which would capture whether frequent care raises spending more for higher-income individuals, is not significant. Specification tests indicate that the linear model misses some curvature and that inference should allow for possible heteroskedasticity.

Section 2 describes the data, Section 3 the empirical strategy, Section 4 the estimates, and Section 5 the specification tests. Section 6 concludes.

## 2 Data

The data come from the IPUMS Health Surveys, a harmonized collection of U.S. health microdata. Each observation is an individual described by 15 variables. The categorical variables are gender, marital status, race, education, and five access and coverage indicators: has a usual place for medical care, has health insurance, has private health insurance, covered by Medicaid, and covered by Medicare. The numerical variables are age, annual income, total medical expenses, and the numbers of doctor visits, emergency room (ER) visits, and nights spent in the hospital.

In cleaning the data, I dropped the identifier column and removed the 150 individuals who reported zero annual income, 6.9 percent of the original 2,166 observations. These are largely students and retirees without reportable earnings, and keeping them would anchor the income distribution at zero in a way that distorts the income-based comparisons this study is built on. The working sample has 2,016 individuals: 933 men and 1,083 women, of whom 41.8 percent are currently married. There are no minors; 1,562 individuals are aged 18 to 64 and 454 are 65 or older. The sample is 68.2 percent White, 21.3 percent Black, 7.3 percent Asian, 2.5 percent multiracial, and 0.7 percent Native American. Educational attainment ranges from less than high school (18.5 percent) through high school (27.2 percent), some college (21.0 percent), and a bachelor's degree (16.6 percent) to a graduate degree (16.7 percent).

**Table 1.** Summary Statistics

| Variable | Min | Max | Median | Mean | SD |
|---|---|---|---|---|---|
| Age (years) | 18 | 99 | 51.0 | 50.68 | 16.73 |
| Annual income ($) | 8 | 260,182 | 26,913.5 | 38,632.70 | 37,565.19 |
| Total medical expenses ($) | 0 | 423,121 | 1,469.0 | 6,924.11 | 19,116.14 |
| Doctor visits | 0 | 195 | 2.0 | 4.18 | 7.42 |
| ER visits | 0 | 8 | 0.0 | 0.24 | 0.68 |
| Nights in hospital | 0 | 44 | 0.0 | 0.38 | 2.16 |

*Notes:* $N = 2{,}016$. The sample excludes 150 individuals with zero reported annual income; income and medical expenses are in U.S. dollars.
*Source:* IPUMS Health Surveys.

Table 1 summarizes the numerical variables. Annual income and total medical expenses both have means far above their medians, the signature of heavy right skew, and the standard deviation of medical expenses ($19,116) is nearly three times its mean ($6,924), reflecting a small number of very high-cost individuals. The histograms in Figure 1 show the same pattern: most people cluster at modest values, with long right tails of outliers, as is typical of income and health care spending.

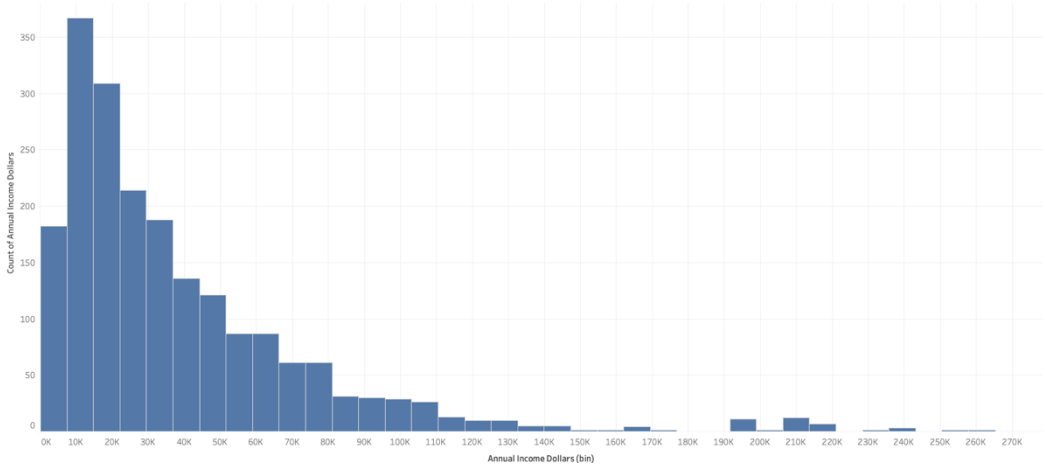
## 3 Empirical Strategy

I estimate three linear models by ordinary least squares (OLS). The first regresses total medical expenses on annual income alone:

$$\text{MedExp}_i = \beta_0 + \beta_1 \,\text{Income}_i + \varepsilon_i \tag{1}$$

The second adds utilization and demographic controls, together with an interaction between income and doctor visits:

*Panel A. Annual income*




*Panel B. Total medical expenses*

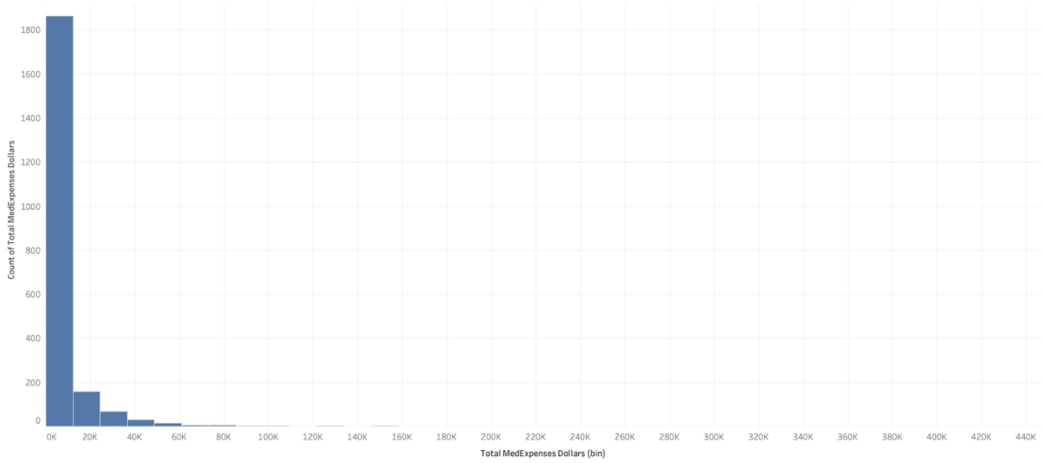



**Figure 1.** Distributions of Annual Income and Total Medical Expenses
*Note:* Histograms for the working sample ($N = 2{,}016$). Both variables are in U.S. dollars; the vertical axes show counts of individuals.

$$\begin{aligned}\text{MedExp}_i = \beta_0 + \beta_1\,\text{Income}_i + \beta_2\,\text{DocVisits}_i + \beta_3\,(\text{Income}_i \times \text{DocVisits}_i) + \beta_4\,\text{Age}_i \\ + \beta_5\,\text{Married}_i + \beta_6\,\text{Male}_i + \beta_7\,\text{Insured}_i + \mathbf{Educ}_i'\boldsymbol{\gamma} + \mathbf{Race}_i'\boldsymbol{\delta} + \varepsilon_i\end{aligned} \tag{2}$$

where $\mathbf{Educ}_i$ and $\mathbf{Race}_i$ are vectors of category indicators. The reference categories are female, not currently married, no health insurance, a bachelor's degree, and Asian. The interaction lets the effect of income vary with the number of doctor visits, and vice versa: the marginal effect of a doctor visit is $\beta_2 + \beta_3\,\text{Income}_i$, and that of income is $\beta_1 + \beta_3\,\text{DocVisits}_i$. If higher-income individuals who see the doctor often spend disproportionately more than either variable alone would predict, $\beta_3$ should be positive. The third model, the baseline specification, is equation (2) without the interaction ($\beta_3 = 0$).

I test the baseline specification in two ways. Ramsey's RESET test adds the squared, cubed, and fourth powers of its fitted values ($\hat{y}^2$, $\hat{y}^3$, $\hat{y}^4$) as regressors and tests whether their coefficients are jointly zero; rejection means that the conditional mean is not linear in the regressors. For

heteroskedasticity, I plot the baseline residuals against income and doctor visits and run a Breusch-Pagan test, regressing the squared residuals on the baseline regressors,

$$\hat{\varepsilon}_i^2 = \alpha_0 + \alpha_1 \,\text{Income}_i + \alpha_2 \,\text{DocVisits}_i + \alpha_3 \,\text{Age}_i + \cdots + v_i \tag{3}$$

and testing the joint significance of the slope coefficients with an $F$-test. All models are estimated in R with `lm()`.

## 4 Results

### *4.1 Income alone*

Column 1 of Table 2 reports the income-only model. The coefficient on income is $-0.0337$: a \$1,000 increase in annual income is associated with about \$34 less in total medical expenses. The estimate is statistically significant ($t = -2.98$, $p = 0.003$), and the $F$-test rejects the null hypothesis of no relationship ($F = 8.88$). Figure 2 shows the fitted line, which slopes slightly downward.

The effect is statistically detectable but economically small. The $R^2$ of 0.0044 means that income explains less than half of one percent of the variance in medical expenses, and the residual standard error of \$19,079 dwarfs the median expense of \$1,469. On its own, income is close to useless as a predictor. The negative sign also runs against the starting hypothesis. A plausible explanation is that income partly captures unobserved health status: lower-income individuals in this sample may be sicker on average and therefore consume more care.

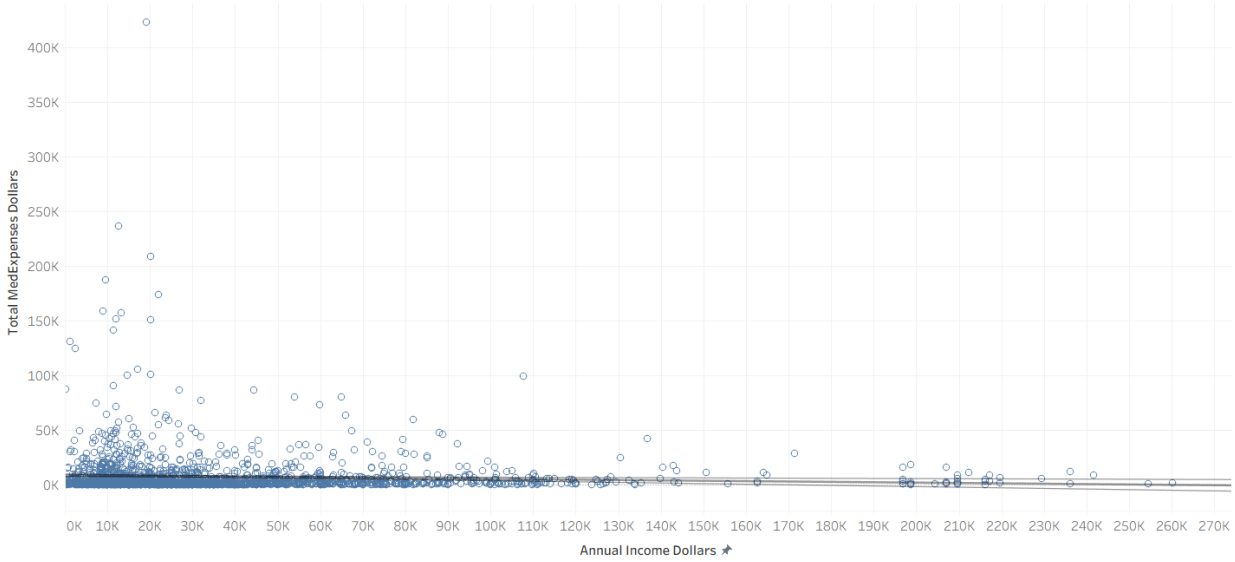



**Figure 2.** Total Medical Expenses and Annual Income

*Note:* Each point is one individual ($N = 2{,}016$). The lines show the fitted simple linear regression (column 1 of Table 2) and its confidence bands. The largest expenses occur at low incomes.

### *4.2 Adding controls and an interaction*

Column 2 adds the controls and the interaction. The number of doctor visits is the strongest predictor. Its marginal effect depends on income: one more visit is associated with about \$945 in additional expenses at an income of \$30,000 and about \$976 at \$60,000. Age comes second:

**Table 2.** OLS Estimates of Total Annual Medical Expenses

| | (1) Income only | (2) With interaction | (3) Baseline |
|---|---|---|---|
| Annual income | $-0.0337^{***}$ | $-0.0294^{**}$ | $-0.0244^{**}$ |
| | (0.0113) | (0.0137) | (0.0115) |
| Doctor visits | | $914.10^{***}$ | $938.59^{***}$ |
| | | (66.02) | (55.16) |
| Income × doctor visits | | 0.00104 | |
| | | (0.00154) | |
| Age | | $93.92^{***}$ | $95.37^{***}$ |
| | | (25.00) | (24.90) |
| Currently married | | $-1{,}564.24^{*}$ | $-1{,}572.99^{*}$ |
| | | (825.81) | (825.59) |
| Male | | 389.87 | 352.74 |
| | | (810.86) | (808.89) |
| Has health insurance | | $2{,}638.63^{*}$ | $2{,}603.10^{*}$ |
| | | (1,359.66) | (1,358.46) |
| *Education* (reference: bachelor's degree) | | | |
| Graduate degree | | 476.95 | 492.68 |
| | | (1,359.41) | (1,359.02) |
| High school | | 420.81 | 422.19 |
| | | (1,263.51) | (1,263.34) |
| Less than high school | | $2{,}392.30^{*}$ | $2{,}397.61^{*}$ |
| | | (1,399.05) | (1,398.83) |
| Some college | | 1,174.26 | 1,178.80 |
| | | (1,314.74) | (1,314.55) |
| *Race* (reference: Asian) | | | |
| Black | | $-1{,}717.99$ | $-1{,}648.74$ |
| | | (1,719.35) | (1,716.06) |
| Multiracial | | $-1{,}593.19$ | $-1{,}548.82$ |
| | | (2,864.72) | (2,863.58) |
| Native American | | 713.09 | 764.11 |
| | | (4,916.15) | (4,914.90) |
| White | | $-589.56$ | $-529.15$ |
| | | (1,542.89) | (1,540.08) |
| Constant | $8{,}226.55^{***}$ | $-2{,}669.12$ | $-2{,}887.09$ |
| | (609.60) | (2,409.11) | (2,387.04) |
| Observations | 2,016 | 2,016 | 2,016 |
| $R^2$ | 0.0044 | 0.1704 | 0.1702 |
| Adjusted $R^2$ | 0.0039 | 0.1642 | 0.1644 |
| Residual std. error | 19,079 | 17,476 | 17,474 |
| $F$-statistic | 8.88 | 27.39 | 29.32 |

*Notes:* The dependent variable is total annual medical expenses in U.S. dollars. Standard errors are in parentheses. In column 2, the coefficient on doctor visits is its marginal effect at zero income. All three $F$-statistics are significant at the 1 percent level.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

each additional year is associated with about \$94 more in expenses ($p < 0.001$), holding the other variables constant.

Income remains negative and significant at the 5 percent level ($p = 0.032$) but small. At 5 doctor visits, a \$1,000 increase in income is associated with about \$24 less in expenses; at 10 visits, about \$19 less. The interaction coefficient is positive, so more frequent visits slightly weaken the negative income effect, but it is far from significant ($p = 0.50$). The data therefore do not support the idea that income and utilization jointly amplify spending.

Three variables are significant only at the 10 percent level. Married individuals spend about \$1,564 less than unmarried individuals ($p = 0.058$); those with less than a high school education spend about \$2,392 more than bachelor's degree holders ($p = 0.087$); and insured individuals spend about \$2,639 more than the uninsured ($p = 0.052$), consistent with the insured using care rather than forgoing it. Gender and the race indicators are not significant at any conventional level. The regressors are jointly significant ($F = 27.39$, $p < 0.001$), but the $R^2$ of 0.170 leaves 83 percent of the variance in medical expenses unexplained.

### *4.3 Baseline specification*

Because the interaction is insignificant, column 3 drops it. The core estimates barely change. Each doctor visit is associated with about \$939 in additional expenses, estimated more precisely ($t = 17.02$) without the collinear interaction term; each year of age with about \$95; and each \$1,000 of income with about \$24 less ($p = 0.035$). The coefficients on marital status, less-than-high-school education, and insurance remain marginally significant, and the $R^2$ is unchanged at 0.170. The baseline retains essentially all of the information in column 2, so I use it for the specification tests.

## 5 Specification Tests

**Table 3.** Specification Tests for the Baseline Model

| | Coefficient | Std. error | $t$ | $p$-value |
|---|---|---|---|---|
| *Panel A. RESET test* | | | | |
| $\hat{y}^2$ | $1.638 \times 10^{-5}$ | $1.782 \times 10^{-5}$ | 0.92 | 0.358 |
| $\hat{y}^3$ | $-3.870 \times 10^{-10}$ | $3.310 \times 10^{-10}$ | $-1.17$ | 0.242 |
| $\hat{y}^4$ | $1.545 \times 10^{-15}$ | $1.314 \times 10^{-15}$ | 1.18 | 0.240 |
| Joint $F$-test | $F(3, 1{,}998) = 6.40, \quad p < 0.001$ | | | |
| *Panel B. Breusch-Pagan test* | | | | |
| Joint $F$-test | $F(14, 2{,}001) = 1.53, \quad p = 0.093$ | | | |

*Notes:* Both tests use the baseline specification (column 3 of Table 2), $N = 2{,}016$. Panel A adds the second, third, and fourth powers of the baseline fitted values to the baseline regression; the coefficients on the baseline regressors are not shown. Panel B regresses the squared baseline residuals on the baseline regressors and tests the joint significance of the slopes. The null hypotheses are a correctly specified linear model (Panel A) and homoskedastic errors (Panel B).

### *5.1 Functional form*

Panel A of Table 3 reports the RESET test. None of the three higher-order terms is individually significant, a typical symptom of the strong collinearity among powers of the same fitted values. Jointly, however, they are highly significant ($F = 6.40$ with 3 and 1,998 degrees of freedom, $p < 0.001$),

so linearity is rejected: the relationship between medical expenses and the regressors has curvature that the linear specification misses. Likely sources are age, where life-cycle effects may make the relationship concave or convex; doctor visits, where costs per visit may diminish or accelerate; and income, for which log transformations are standard in this literature.

### *5.2 Heteroskedasticity*

Figure 3 plots the baseline residuals against income and doctor visits. Against income, the residuals are not clearly fan-shaped, although their spread is larger at low incomes, where most of the sample lies. Against doctor visits, the pattern is more striking: most residuals stay close to zero at low visit counts, while larger positive and negative residuals appear at higher counts, a visual warning sign of heteroskedasticity.

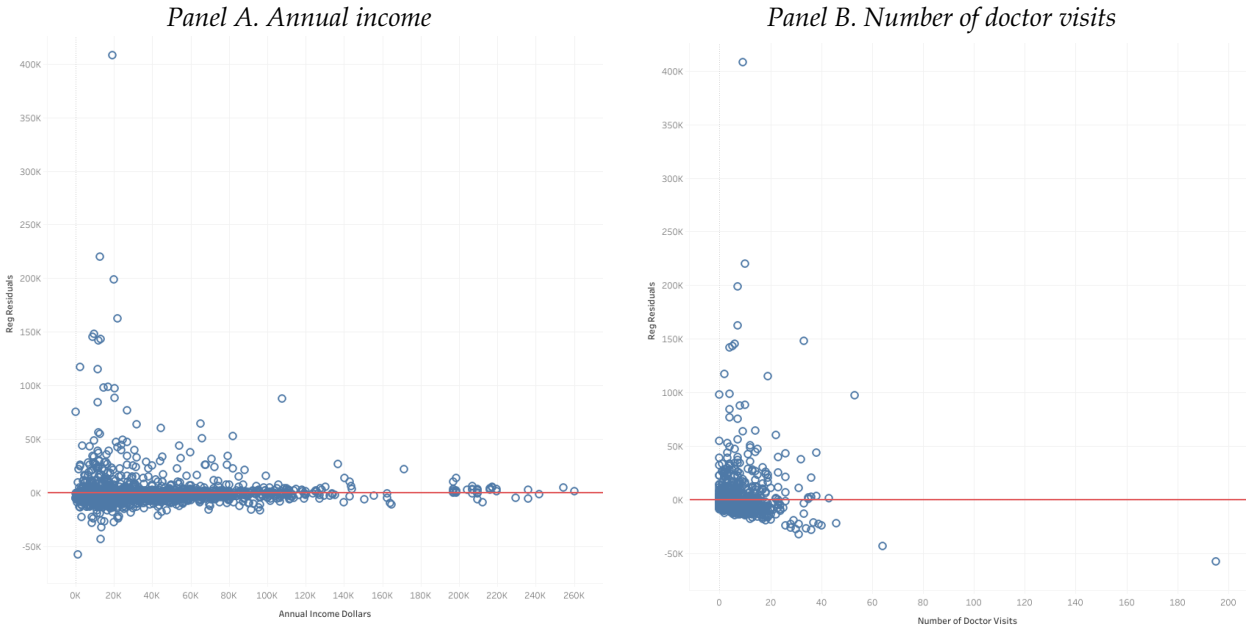



**Figure 3.** Baseline Residuals against Annual Income and Doctor Visits

*Note:* Residuals from the baseline specification (column 3 of Table 2). The red line marks zero.

The Breusch-Pagan test (Table 3, Panel B) gives $F = 1.53$ with 14 and 2,001 degrees of freedom ($p = 0.093$). The test fails to reject homoskedasticity at the 5 percent level, although it would reject at the 10 percent level, and the doctor-visits plot points toward non-constant variance. The evidence is therefore mixed: the formal test does not flag heteroskedasticity at the 5 percent level, but the residuals are not unambiguously homoskedastic either. Heteroskedasticity-robust (HC1 or HC3) standard errors, or weighted least squares with weights based on an estimate of the error variance as a function of doctor visits, would protect inference against any remaining heteroskedasticity.

## 6 Conclusion

Within the variables available, the determinants of medical spending are simple and intuitive: how often people see the doctor and how old they are dominate. In the baseline specification, each additional doctor visit is associated with roughly $940 more in annual medical expenses, and each additional year of age with roughly $95 more. Income, the variable that motivated the study, plays a minor role, and its negative sign is likely absorbing unobserved health status, so it should not be

read as a causal effect. The interaction between income and doctor visits is not significant; marital status, less-than-high-school education, and insurance status are significant only at the 10 percent level; and gender and race are not significant.

The $R^2$ of 0.17 is arguably the most informative statistic in the analysis. It implies that most of the variation in medical spending is driven by factors this dataset does not measure, such as specific chronic conditions, prescription drug regimens, the generosity of a person's insurance plan, provider networks, and geography. The specification tests add two caveats: the RESET test rejects the linear functional form, and the residuals may be heteroskedastic.

Four extensions follow directly. First, re-estimating with log medical expenses or log income would address the right skew and test whether elasticity-style specifications fit better. Second, polynomial terms in age and doctor visits would address the RESET result. Third, inference should be checked with HC1 or HC3 robust standard errors or by weighted least squares. Fourth, a two-part model, with a logit for whether any spending occurs and a log-linear model for spending among those with positive expenses, is standard in the health care expenditure literature.

## Data and Code

Data: IPUMS Health Surveys. Estimation: R, using `lm()` for OLS. Diagnostics: Ramsey RESET test (powers 2 through 4 of the fitted values) and Breusch-Pagan test on squared residuals.